

SWARNIMA SINGH

B.Tech Student in Artificial Intelligence & Machine Learning

Email: rajputsinghswarnima@gmail.com GitHub: github.com/Swarnima-web Portfolio: swarnima-portfolio-phi.vercel.app



PROFILE SUMMARY

Passionate Artificial Intelligence and Machine Learning engineering student at Bennett University with a strong builder's instinct. Driven by a deep curiosity for intelligent systems, robotics, and interactive environments. Analytical thinker and dedicated problem solver, skilled at translating theory into functional repositories, debugging workflows, and building structured code. Continuous learner exploring advanced AI and game engineering paradigms.

ACADEMIC PROJECTS

AI Learning Lab

Python • Jupyter Notebooks

A hands-on workspace compiling machine learning and mathematical algorithms.

- Designed model training workflows focusing on foundational statistics and data representations.
- Implemented clear, reproducible notebooks covering classification and optimization models.
- Iteratively tested and documented algorithm outputs to establish solid machine learning pipelines.

Game Systems Prototype

HTML5 Canvas • JavaScript

Interactive web-based gameplay environment focusing on client-side state execution.

- Built a lightweight, responsive game loop executing real-time coordinate physics and collision logic.
- Designed clean UI structures for game states, player triggers, and feedback animations.
- Managed event listeners and state machines to optimize player interactions and framerate stability.

Robotics Notes Engine

Python • Markdown

A repository consolidating notes, motion simulation templates, and emerging hardware documentation.

- Compiled engineering concepts connecting matrix computations to physical kinematics models.
- Structured system dynamics notes mapping sensor integration schemas and signal flows.
- Created a modular markdown engine to navigate robotic architectures and algorithm paradigms.

CERTIFICATIONS

AI Python for Beginners

DeepLearning.AI

Engineering Probability and Statistics (Parts 1 & 2)

Northeastern University

Solving Ordinary Differential Equations with MATLAB

MathWorks

Solving Nonlinear Equations with MATLAB

MathWorks

EDUCATION

Bennett University

B.Tech in AI & ML
Greater Noida, Uttar Pradesh, India

2025 – 2029

Om National Academy

+2 (Science Stream)
Birgunj, Nepal (Bio + Math Focus)

2023 – 2025

Kadambari Academy

Class X Secondary Ed.
Birgunj, Nepal

2013 – 2023

TECHNICAL SKILLS

Programming Languages

Python, JavaScript, MATLAB, HTML5, CSS3, C, C++

Artificial Intelligence & ML

Model workflow, Probability & Stats, ML Notebook logic

Game Development

Game loops, coordinate physics, interactive gameplay systems

Robotics Engineering

Kinematics concepts, sensor workflows, system logic

Analytical Strengths

System decomposition, debugging workflows, modular logic

CORE HOBBIES

Gaming, Game Prototyping, Coding experiments, Sensors & Robotics, Solving algorithmic puzzles, Exploring emerging technologies.

LANGUAGES

English Hindi Nepali